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A deep learning approach to predict significant wave height using long short-term memory

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ABSTRACT

We present a new deep learning training framework for forecasting significant wave height on the Southwestern Atlantic Ocean. We use the long short-term memory algorithm (LSTM), trained with the ERA5 dataset and also with buoy data. The forecasts are made for seven different locations in the Brazilian coast, where buoy data are available. We consider four different lead times, e.g., 6, 12, 18 and 24 h. Experiments are conducted using exclusively historical series at the selected locations. The influence of other variables as inputs for training is investigated. Results of the LSTM forecast show that a data-driven methodology can be used as a surrogate to the computational expensive physical models and also as an alternative to the reanalysis data. Accuracy of the forecasted significant wave height is close to 87% when compared to real buoy data.

1. Introduction

Almost all engineering applications concerning the ocean, from navigation to renewable energy, going through offshore platforms, alerts of catastrophic events and geosciences research (Komen et al., 1996; Cavaleri et al., 2007; Ardhuin et al., 2019), benefit from an accurate description of the sea state, of which wave heights are probably the most important parameter.

It is no novelty that ocean waves can be simulated with mathematical–physical models, and several state-of-the-art programs are available for such purpose (see, e.g., the WAVEWATCH III WAVEWATCH III Development Group (WW3DG), 2019 and SWAN Booij et al., 1997). However, powerful artificial intelligence algorithms are gaining visibility and popularity with the increasing public computational libraries available making it possible the analysis of large amount of data for several applications that are based on historical information. As several reliable databases are available and continue to expand the time period for which they provide historical data, one important question has been posed: can data-driven models, with help of the artificial intelligence, act as a physical model surrogate, with computational time and accuracy that are superior to the latter? This question has already been posed and a compeling review can be found in Boukabara et al. (2019, 2022) and Boukabara and Hoffmanb (2022).

The use of artificial intelligence, specially in the context of artificial neural networks (ANNs) in wave modelling has already been investigated and studied. In a pioneering research, Deo and Naidu (1998) used feed-forward networks with three different training approaches to forecast wave heights along the east coast of India. They used buoy observations data as input to the model, collected every three hour, from May 1983 until August 1984, resulting in 16 months of historical data. Different lead times were analysed and the predictions yielded accurate results. From this point on, ANNs have been used in several works for wave predictions.

Short-term forecasts using ANNs trained with data from two sites offshore the Atlantic and the Irish Sea coasts of Ireland is reported in Makarynskyy (Makarynskyy, 2004), and hourly forecasts of significant wave height and zero-up-crossing wave periods are made for 1–24 h time intervals. In another ANNs work, Makarynskyy and collaborators (Makarynskyy et al., 2005) have used buoy data from Portugal's West coast, also to forecast significant wave height and zero-up-crossing wave period for 3, 6, 12 and 24 h intervals. In one of the approaches, each parameter over every time interval is forecasted by a separate ANN, while in the second approach, only two ANNs are used to concurrently simulate the variables. Browne et al. gave a comprehensive explanation of using ANNs to estimate waves nearshore and compared results with SWAN model. Results show that ANNs outperformed the physical model in simulations for seventeen nearshore locations around the continent of Australia over a 7 month period.

Agrawal and Deo (2004) applied ANNs as an alternative to find interrelationships among certain characteristic wave parameters. Networks were trained for locations at the east coast of India and developed in order to estimate values of average zero-cross wave period,

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peak-spectral period, maximum spectral energy density and maximum wave height from the given value of significant wave height and also to evaluate the spectral width parameter from the spectral narrowness parameter. Krasnopolsky et al. (2002) proposed an alternative to the complex mathematical formulations involved in forecast systems by approximating solutions of exact physical models using ANNs. They consider the UNESCO equation of state of the seawater (density of the seawater) and an approximation for the nonlinear wave–wave interaction. The non-linear interactions in wind wave spectra is also investigated by Tolman et al. (2005) using ANNs, while Zamani et al. (2008) forecasted significant wave heights for several hours ahead using buoy measurements in Caspian Sea using models based on ANNs.

Londhe and Panchang (2006) applied ANNs within the feed-forward back propagation algorithm for four different lead times (6, 12, 18 and 24) to forecast significant wave heights in six different buoy locations at the Gulfs of Mexico, Alaska and Maine. Six network architectures were tested for each buoy location, and predictions were made for the period of 8 January until 31 December, 2004 for all buoys except one, for which the predictions ended in 16 September of that year. An accuracy of 86% is obtained for the 6 h lead time and between 67% and 83% for 12 h. Besides that point, as expected, the accuracy drops to 55%–71% for 18 h lead time and less than 63% for the 24 h lead time forecast. Despite the good results, a limitation highlighted by using ANNs in ocean wave predictions is the considerable under-predictions in the highest peaks, which can be explained by the dominance of smaller wave heights records in the datasets used to train the network. In an attempt to overcome this drawback, data from 2004 were added to the training phase (which had records of the Hurricane Ivan), and a tendency to catch the higher peaks was observed.

More recently, Campos et al. (2020, 2017) developed a post-processing algorithm to improve ensemble averaging, as a replacement to the typical arithmetic ensemble mean, using neural networks trained with altimeter data. Similar techniques using nonlinear ensemble averaging, also based on neural networks, were studied in the Gulf of Mexico (Campos et al., 2019). James et al. (2018) used supervised machine learning algorithms to estimate ocean-wave conditions at the Monterey Bay, California, to act as a surrogate to physical models, while O'Donncha et al. (2018) combined both ensemble physical model simulations and machine learning to derive an integrated technique that investigate uncertainty from modelling and generate a forecast that is better than the best individual model prediction. Bento et al. (2021) used Convolutional Neural Networks (CNN) to forecast short-term wave power in different seasons of the year, with results that outperforms conventional methods for horizons between two and six hours.

Not only ANNs were used to ocean wave predictions in the context of data-driven results, but also several other artificial intelligence techniques, such as support vector machines (Browne et al., 2007), Bayesian optimization (Cornejo-Bueno et al., 2018), genetic programming (Nitsure et al., 2012; Gaur and Deo, 2008) and wavelets (Oh and Suh, 2018; Prahlada and Deka, 2015). Furthermore, Deep learning, i.e., neural networks within a deep range that uses great amount of data, has already been shown to be a powerful technique for oceanography predictions (Zheng et al., 2020; Choi et al., 2020).

The traditional architecture of ANNs had already been improved. If the network has at least one feedback loop, forming a cycled connection between layers, then we create recurrent neural networks (RNNs), an architecture that aims to give a better accuracy on predictions that needs a memory along the network. Nevertheless, the use of RNNs in its standard configuration to account for contextual information is still limited, due to the effect known as vanishing gradient problem (Hochreiter et al., 2001). As the information circles around the recurrent network in time, the influence of a input on the hidden layer, and consequently on the output, either decays or blows up exponentially. One attempt to solve this problem is with the Long Short-Term Memory (LSTM) architecture, presented for the first time by Hochreiter and Schmidhuber (1997).

The long short-term memory (LSTM) recurrent neural network has the ability of learning long-term dependencies. This benefit gives the possibility to create a neural network that can use information of a long past to build its predictions. LSTM architecture presents a 'forget gate', a modification made to overcome the vanishing gradient problem. In a recent work, Pirhooshyaran and Snyder (2020) used LSTM together with a sequence-to-sequence neural network to forecast and hindcast ocean waves, as well as to reconstruct missing data. Besides, feature selection has been employed based on nearby buoys data. Hu et al. used LSTM to predict ocean wave height and period under the near-idealized wave growth conditions of Lake Erie (Hu et al., 2021), obtaining an accuracy close to numerical models but reduced storm peak underestimation. Lou et al. (2021) proposed a framework for automatic driving scheme combining wave height prediction using LSTM with ship driving in order to adjust ships course and avoid areas with higher waves. Predictions of wave height, wave period, and wave direction in the US Atlantic Coast using an ANN constructed with both convolutional LSTM and LSTM layers is conducted by Wei (2021). The ANN is used to forecast storm waves induced by winter storms, Hurricanes Isaias and Eta with good accuracy specially for short lead times.

In the context of ocean wave modelling, given the comprehensive amount of significant wave height time series documented and available, LSTM is a good alternative to improve predictions of this variable.

Thus, the present work aims to develop a new training methodology for LSTM neural networks to forecast significant wave height on several locations in Brazil's coast, for four different lead times. We consider only the historic wave data to feed the net since, as it will be shown, additional features show no improvement in the final results. Our approach differs from the work in Pirhooshyaran and Snyder (2020) both in methodology and goals. We choose to perform forecast with verification times within a time period of one month (or 744 time-steps) and the training phase of the LSTM considers four different lead times, namely, 6, 12, 18 and 24 h. Moreover, we solely focus on LSTM predictions for significant wave height, using (i) only this variable for uni-variate forecasts; (ii) closely correlated variables, such as peak wave period and 10m wind speed and (iii) the best four variables based on Pearson's correlation coefficient, for multi-variate forecast. Training of the LSTM network is performed using both ERA5 reanalysis data, implemented by the European Center for Medium Range Forecast (ECMWF) (Hersbach et al., 2018, 2020) and real buoy measurements. As our contribution, we present a new methodology to train LSTM network that can be suitable for short-term forecasts of significant wave height, and potentially for other ocean waves variables as well, with a large decrease in computational time if compared to traditional physical models and with improved accuracy. Moreover, results using ERA5 as training dataset does not aim to provide an operational forecast, because these data are usually not available on a daily basis, but training the LSTM with buoy dataset is an independent experiment and could be adapted to be used as a possible operational forecast.

This paper is structured as follows: In Section 2, we show the mathematical and theoretical background of the machine learning algorithm used in this work, the LSTM . In Section 3, the data used are described and in Section 4 the methodology of our framework is presented. Section 5 shows the predictions results based on LSTM and comparisons with ERA5 and buoy observations. Finally, the conclusions are presented in Section 6.

2. Artificial neural networks

Machine learning algorithms are capable of producing data-driven decisions. Supervised learning is a type of machine learning where both input and output data are given in the training phase. The algorithm learns the mapping function from the input to output variables, and after that, with a new set of data, the algorithm produces an output.

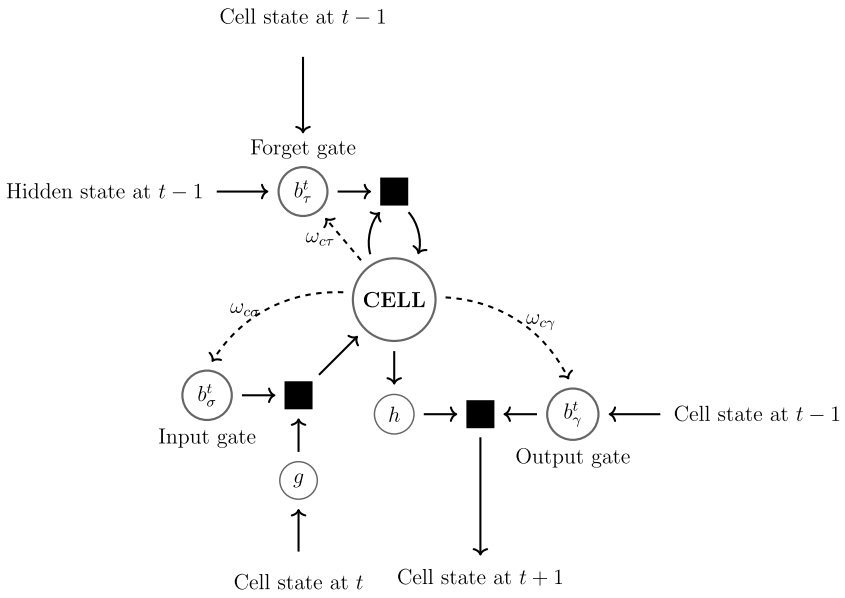


Fig. 1. LSTM memory block. The input, output and forget gates that controls the activation of the cell through the multiplicative units (black squares) from inside and outside. Note that the forget gates will multiply the previous state of the cell. Functions g and h are activation functions explained in the text.

Amongst machine learning techniques, artificial neural networks (ANN) were developed motivated by the way in which the brain performs a particular task or function of interest (Haykin, 2009). From a mathematical standpoint, they can be considered as multiple non-linear regression methods able to capture hidden complex nonlinear relationships between input and output variables (Peres et al., 2015).

Several ANNs architectures that are based on layers of the neurons are possible. A feedforward network consists of a connection between input layer (where the data are gathered), hidden layers and the output layer (which gives us our desired result). The term 'hidden' refers to the fact that this part of the network is not seen directly either by the input and the output layers (Haykin, 2009). We can consider a single-layer architecture, where no hidden layers are used. The benefit of hidden layers is to give the network a global perspective due to the extra set of synaptic connections and dimension of neural interaction (Churchland and Sejnowski, 1994).

Differently, recurrent neural networks store some information about the past time evolution of the system in a hidden state vector. This means that a neuron's output can be feedback as an input to all neurons of the net. A self-feedback network occurs when an output neuron feeds its own input, while a no self-back network means otherwise, that is, the output of one neuron is used as input to all other neurons but itself. This attribute of RNNs allows a memory of previous inputs to persist in the network's internal state, and thereby influence the network output (Graves, 2008). RNNs can be derived from nonlinear first-order non-homogeneous ordinary differential equations; a deep and elucidate analysis can be found in Sherstinsky (2020). Nevertheless, the use of RNNs in its standard configuration to account for contextual information is still limited, due to the vanishing gradient problem (Hochreiter et al., 2001).

2.1. Long short-term memory (LSTM)

With the purpose of solving the vanishing gradient problem, long short-term memory (LSTM) architecture incorporates non-linear, data-dependent controls into the RNN cell, so that the gradient of the loss function does not vanish (Sherstinsky, 2020). One difference between LSTMs and RNNs is that the summation in the hidden layer is replaced by a memory block, which has four neural network connected and interacting together. This structure allows LSTMs to learn and remember information for a long-time period, which is its default behaviour.

The architecture of LSTMs is build as follow: inside the memory block, there is one or more central cells that are self-looped into three multiplicative units called input, output and the forget gates. This difference of having more units controls the flow of information (Goodfellow et al., 2016), where the multiplicative input protects the memory block from receiving perturbation from irrelevant inputs, while the output gates protect other units from irrelevant information of the current block (Hochreiter and Schmidhuber, 1997). The units work as gates to avoid weight conflicts, i.e., the input gate decides when to keep or exclude information within the block, while the output gate decides when to access the block and prevent other blocks from being perturbed by itself.

Fig. 1 shows an overview of the memory block inside a LSTM network. The three gates receive activations from both inside and outside (other memories blocks), controlling the activation of that cell by multiplications. The input and output gates multiply the input and output of the cell while the forget gate multiplies the previous state of the cell. If n, m and k correspond to the number of inputs, outputs and cells in the hidden layer, respectively, the activation of the input gate b_σ^t at time t is given by

$$b_\sigma^t = \phi\left(\sum_{i=1}^{n}\omega_{i\sigma}x_i^t + \sum_{h=1}^{m}\omega_{h\sigma}b_h^{t-1} + \sum_{c=1}^{k}\omega_{c\sigma}s_c^{t-1}\right), \tag{1}$$

where σ represents the input gate, x the signal, ω the weights that will connect two units and s_c^t is the activation of the cell c at time t unit. Usually, the gate activation function ϕ is the logistic sigmoid so that the values are between 0 and 1 (Hochreiter and Schmidhuber, 1997; Graves, 2008; Goodfellow et al., 2016). The activation of the forget gate b_τ^t at time t is given by

$$b_\tau^t = \phi\left(\sum_{i=1}^{n}\omega_{i\tau}x_i^t + \sum_{h=1}^{m}\omega_{h\tau}b_h^{t-1} + \sum_{c=1}^{k}\omega_{c\tau}s_c^{t-1}\right). \tag{2}$$

while the activation of the output gate b_γ^t at time t is

$$b_\gamma^t = \phi\left(\sum_{i=1}^{n}\omega_{i\gamma}x_i^t + \sum_{h=1}^{m}\omega_{h\gamma}b_h^{t-1} + \sum_{c=1}^{k}\omega_{c\gamma}s_c^{t}\right). \tag{3}$$

The activation of the cell s_c^t at time t is

$$s_c^t = b_\tau^t s_c^{t-1} + b_\sigma^t g\left(\sum_{i=1}^{n}\omega_{ic}x_i^t + \sum_{h=1}^{m}\omega_{hc}b_h^{t-1}\right) \tag{4}$$

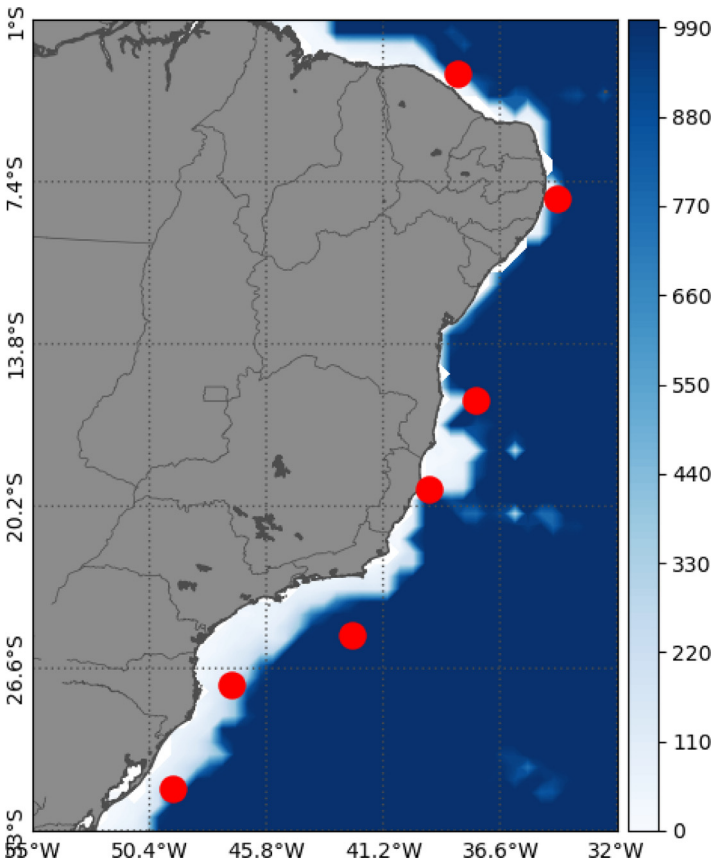


Fig. 2. Region of analysis in the Brazilian coast with bathymetry. Red circles indicates the location of the seven buoys studied in this work. The blue colour scale represents the bathymetry.

where g, the activation function, is usually a hyperbolic tangent of logistic sigmoid functions. Finally, the cell output b_c^t at time t is given by

$$b_c^t = b_\gamma^t h(s_c^t) \tag{5}$$

where h can be the same function as g, or even the identity function. In the formulae above, $\omega_{c\sigma}$, $\omega_{c\tau}$ and $\omega_{c\gamma}$ indicate the weights from the cell to the input, forget and output gates, respectively. Note that if the input gate has an activation near zero, it will not open, and therefore the activation of the cell will not be overwritten by new inputs, and will be available later in the sequence, i.e., the block store information for a longer time than usual (Graves, 2008).

The calculations explained above are provided for the forward pass of a LSTM hidden layer, starting at $t = 1$ and applying the equations iteratively to update t, until the length of the input sequence of data is achieved. However, to complete the calculations, a backpropagation backward pass is necessary, starting at the length of the input and going back until $t = 1$. The formulae of this phase can be found in Hochreiter and Schmidhuber (1997), Graves (2008) and Goodfellow et al. (2016).

3. Data and area of study

The LSTM methodology described in the previous section is used here to predict the significant wave height H_s, based on the ERA5 reanalysis and observational data. Values of H_s are gathered from ERA5 hourly from 1979 to present (Hersbach et al., 2018, 2020). ERA5 is the fifth generation ECMWF reanalysis for the global climate and weather for the past 4 to 7 decades. We chose ERA5 reanalysis dataset since it is an *approximation* of the true sea state, provides comprehensive global data publicly available, it allows any user to test the present method and provides long-term data (in contrast to buoy data). In our experiments, the training dataset is selected until one month prior to the predictions period (see Table 1).

Seven buoys locations are considered for this study. All of them are located in the Brazilian coast, ranging from longitude 49° 86′W to 38° 25′W and latitudes 31° 33′S to 3° 12′S. These buoys belong to the National Program of Buoys (PNBOIA) of the Brazilian Navy, which aims to collect oceanographic and meteorological data of the Atlantic Ocean (Pereira et al., 2017; Navy, 2020). A detailed explanation of data quality can be found in Navy (2019). Fig. 2 shows the region of analysis, with red circles indicating the location of the buoys, while Table 1 presents the longitudes and latitudes of the seven buoys. Predictions are made for verification times within one month, but the month chosen vary for each buoy, due to the lack of complete data in the databases. Table 1 also shows the period of predictions to each buoy location. Nowadays, only buoy 7 is still operating, while all others are in maintenance.

In this work, we use three metrics to analyse the accuracy of our results. The well known mean absolute error (MAE), which is given by

$$MAE = \frac{1}{n}\sum_{i=1}^{n} \left|\tilde{y}_i - y_i\right|, \tag{6}$$

where the tilde means reference value while non-tilde means predicted value (n is the number of observations). The relative error (RE) for each step is given by

$$RE = 100\frac{\left|\tilde{y}_i - y_i\right|}{\left|\tilde{y}_i\right|}, \tag{7}$$

and the mean absolute relative error (MAPE),

$$MAPE = 100\frac{1}{n}\sum_{i=1}^{n} \frac{\left|\tilde{y}_i - y_i\right|}{\left|\tilde{y}_i\right|}. \tag{8}$$

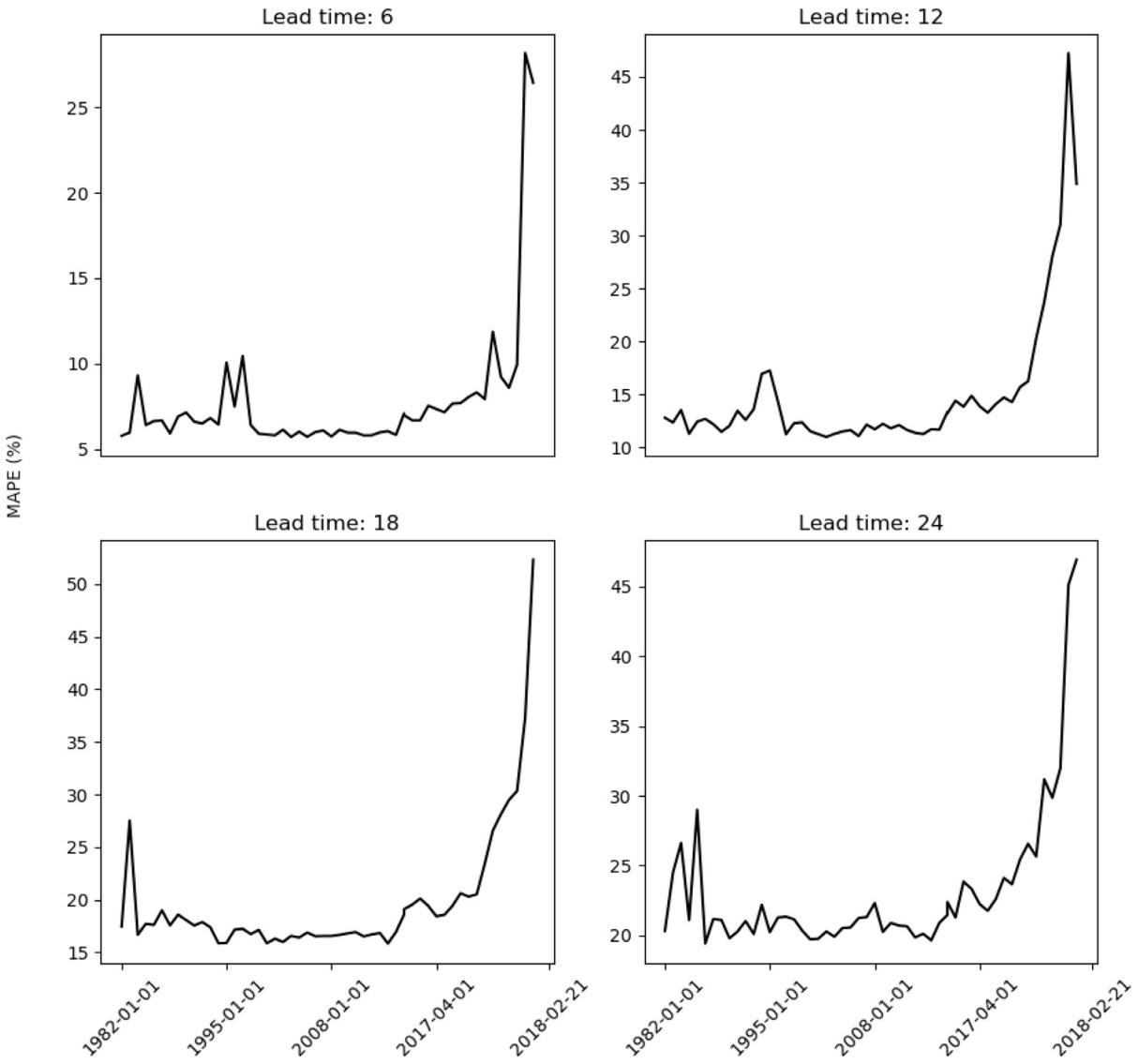


Fig. 3. MAPE calculated for LSTM predictions using different starting dates: from January/1979 until February/2018. Simulations were performed for buoy number 1 location and ERA5 as training dataset.

Table 1
Geo-spatial latitude and longitude location of the seven buoys used in this work, period of prediction, water depth and city of location in Brazil.

	Longitude	Latitude	Period of prediction	Water depth (m)	WMO	City/State location
Buoy 1	49° 86' W	31° 33' S	March/2018	200	31053	Rio Grande/RS
Buoy 2	47° 15' W	27° 24' S	March/2018	200	31231	Itajaí/SC
Buoy 3	42° 44' W	25° 30' S	January/2021	2164	31374	Santos/SP
Buoy 4	39° 41' W	19° 55' S	March/2017	200	31380	Vitória/ES
Buoy 5	37° 56' W	16° 00' S	March/2016	200	31260	Porto Seguro/BA
Buoy 6	34° 33' W	8° 09' S	March/2016	200	31229	Recife/PE
Buoy 7	38° 25' W	3° 12' S	March/2017	200	31229	Fortaleza/RN

Both MAPE and RE are given in percentages while MAE is in the same unit as the data.

Since the training dataset time span can have impact in the predictions, both on accuracy and processing time, we conducted an experiment with the data gathered from 1979 until 2018, to find that, for different lead times, the amount of data needed for a better accuracy is different. We show in Fig. 3 the MAPE between LSTM predicted and ERA5 reanalysis data, for starting the date of training set from 01-01-1979 to 28-02-2018 (see the x-axis), i.e, data range from 39 years to one week prior to the prediction period, which is the same for each starting date (defined in Table 1). Although we are comparing a forecast from LSTM with reanalysis from ERA5, this difference can be used for the purpose of selecting the right dataset size, since training of the network is performed using ERA5 data in this case. Simulations and the establishment of the training dataset to be used adopted exclusively buoy number one (Rio Grande). As we can see, the MAPE starts to grow exponentially with a training dataset of only one year, from February/2017 until the end of February/2018, which is an expected behaviour, considering the need for historical data of the algorithm. However, as the training set becomes larger than one year, we see that although LSTM is developed to remember historical data, it is not that the bigger the dataset, the better is the accuracy. In fact, for all lead times, there is an increase of MAPE in some datasets that are longer than 25 years. This could happen because of the quality of the data

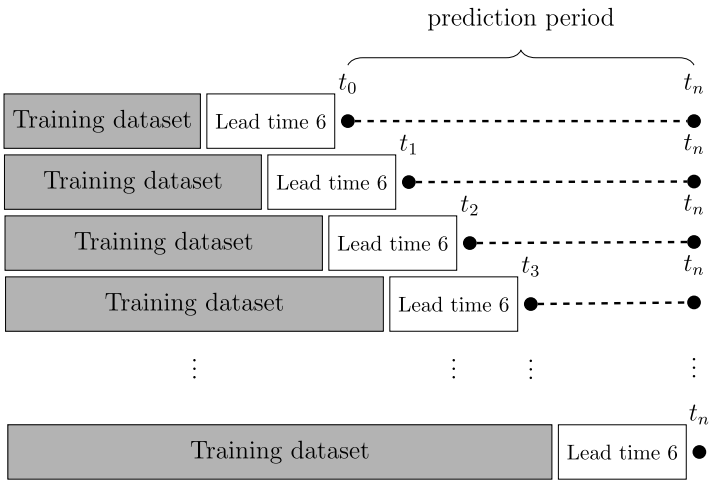


Fig. 4. Schematic of the methodology for training applied in this work, for lead time six (other lead times follow analogously). For each verification time t_i, where $i = 1, \ldots, n$, the training set consists of data until six hours steps prior to t_i.

Table 2
Training set starting date best MAPE accuracy results for four different lead times considered in this work.

Lead	MAPE	Training start date
6	5.70%	01-01-2003
12	10.98%	01-01-2002
18	15.85%	01-01-2005
24	19.41%	01-01-1987

or how the variable is changing over time, and is object for further investigation.

Thus, to reach the better results for a prediction, we consider the accuracy, size of the training set and global accuracy, both by MAPE, MAE and RE. Based on this analysis, we set the starting date of the training sets for each lead time differently, as summarized in Table 2. Still, it is possible to reach good accuracy with one year of dataset; simulations will run fast and the predictions will be satisfactory, although with some loss of accuracy if compared with the training size datasets we have chosen in the work. Its important to mention also that, with smaller datasets, the larger lead time predictions, specially eighteen and twenty four, will fail to predict peaks or valleys since the network will approach the average value of the period.

4. Methodology

The main goal of this work is to predict the significant wave height H_s for the seven buoys locations presented in Table 1, hourly for four lead times (6, 12, 18 and 24), using the LSTM neural network. As mentioned, the starting date of training set is different for each lead time, and is defined as showed in Table 2, while the end of the database depends on the lead time, that is, for each verification time, a new training is performed with the training set consisting of data from the start date until the prediction time minus the lead time. With this, we guarantee that our training will have as much as data necessary for the best result, respecting the lead time gap. Thus, in our methodology, we make an one hour forecast, that is separated from the training dataset by the lead time hours, for each verification time. This methodology differs from the traditional train-predict machine learning framework, since we want to preserve the gap between train and predicted step. Fig. 4 shows a schematic of the training methodology applied.

We considered different strategies for the prediction: first, only the historical series of H_s in the buoy location is used for training. We also investigate the influence of other variables in the training set, based both on physical aspects and on a correlation analysis. Thus, predictions of H_s is also performed using other physical variables, but excluding H_s, as input. All above simulations use ERA5 in the training dataset. Lastly, we perform simulations with real buoy data observations to train the network. A summary of the simulations is presented in Table 3 and explained in more detail in Section 5.

In the training phase of the LSTM, a cross-validation scheme was implemented, where 80% of the data are selected for the training and 20% for validation. This strategy is an excellent framework to avoid overfitting of a model, i.e., a model that yields a good accuracy to the validation set (seen data) and a bad result to unseen data. Thus, in our methodology, the data were divided into the effective training set, the validation set (80/20 split) and the set of values to predict, called test set, using a standard terminology from machine learning algorithms.

To avoid excess in the memory usage of the training phase, the full set of data is subdivided into smaller batches. Training is then performed for each batch, with the target value considered for this batch, and respecting the lead time. To define the batch size, several simulations were performed, and a optimal value of twelve data points was obtained.

The Python library TensorFlow (Abadi et al., 2015) is an end-to-end open source platform for machine learning and its Keras API (Chollet et al., 2015) are used in this work to implement the LSTM algorithm. The model is compiled using the mean absolute error as loss function which is optimized by the Adam algorithm (Kingma and Ba, 2014). Adam optimization is a stochastic gradient descent method that is based on adaptive estimation of first-order and second-order moments. The LSTM network is build with three layers, one with 64, other with 48 neurons and the last with 32. To tune the model, a hyper-parameters optimization was held and these values yields the best results possible, considering the metric (MAPE) and the computational time. The code used in the simulations can be found in Minuzzi (2022).

5. Results

In this section, we present the results of the predictions for each buoy location and lead time. Simulations are performed using both GPU and CPU parallelization, to improve performance. Early stopping is used for convergence of the solution, i.e., as the learning rate achieves a desired accuracy based on MAE, the training stops.

For each lead time, our algorithm took approximately 14 min for training the LSTM and the prediction of a single time value took 2.62×10^{-6} s. The simulations were performed in a machine with Intel Xeon processor with 20 cores, 128 Gb of RAM memory, with a GeForce RTX 2080 Ti graphic card. The GPU is used to parallelize the training process.

Table 3
Summary of simulations developed in this work.

Framework	Features	Target	Training set	Validation set	Test set
Only the historical series of H_s	H_s	H_s	ERA5	20% of training	ERA5
Multivariate predictions of H_s	H_s, T_p, U_{10}	H_s	ERA5	20% of training	ERA5
Multivariate predictions of H_s	Best correlated variables with H_s	H_s	ERA5	20% of training	ERA5
Predictions not considering H_s as feature	T_p and U_{10}	H_s	ERA5	20% of training	ERA5
Predictions using buoy data in the training set	H_s	H_s	Buoy data	20% of training	Buoy data

Table 4
MAPE and MAE metrics for each buoy location and lead times for H_s predictions with LSTM.

	MAPE				MAE			
	6	12	18	24	6	12	18	24
Buoy 1	6.12%	11.9%	18.82%	24.15%	0.13 m	0.26 m	0.39 m	0.50 m
Buoy 2	4.88%	9.36%	12.98%	14.69%	0.08 m	0.16 m	0.23 m	0.26 m
Buoy 3	4.28%	9.06%	11.83%	13.26%	0.08 m	0.17 m	0.23 m	0.25 m
Buoy 4	4.67%	8.87%	11.34%	12.27%	0.06 m	0.12 m	0.15 m	0.17 m
Buoy 5	3.15%	6.27%	6.64%	8.25%	0.04 m	0.08 m	0.08 m	0.11 m
Buoy 6	2.19%	3.7%	4.97%	5.98%	0.03 m	0.05 m	0.07 m	0.09 m
Buoy 7	2.75%	4.54%	4.71%	6.2%	0.04 m	0.07 m	0.08 m	0.10 m

5.1. Predictions using H_s both as feature and target

In this section, we show the results of the LSTM forecast, using ERA5 as training dataset and only H_s as feature and target. All metrics here are calculated using ERA5 as reference value in the formulas (6)–(8). Figs. 5–7 present the comparison between predictions obtained with LSTM, the ERA5 reanalysis (same data in the four figures) and the observations, for buoys locations number two, five and seven, respectively. As it can be observed, lead time 6 yields a lower MAPE, calculated between LSTM prediction and ERA5 reanalysis data, in all the locations, which is expected since the gap between the final step of training and the predictions step is smaller, so the memory that the network needs to keep is smaller. For buoy location number two, based on the MAPE, the LSTM results show an accuracy[1] of 95.12% for lead time 6, while accuracy is 90.64% for lead time 12, accuracy of 87.02% for lead time 18 and accuracy of 85.31% for lead time 24. We can see that the pattern for lead time 6 is very well described, with the forecast series closely following the reanalysis values and the buoy data. Except for two times, e.g., 2018-03-20 at 1:00:00 UTC and 2018-03-20 at 2:00:00 UTC, where the relative error is above 30%, all errors are below 25% for lead time 6 whilst the MAE is equal to a remarkable 0.086 m. Despite a small loss of accuracy for lead times 12, 18 and 24, we still observe an accuracy of more than 85% for all lead times in this location.

Predictions for buoys locations number five and seven show the best accuracies in all studied buoys locations, as we can see in Table 4. An outstanding accuracy of 97.25% is obtained for lead time 6 in buoy location number seven, while the 24 h lead time prediction, expected to behave somewhat poorly, has an accuracy of 93.8%. The predicted values follows closely the reanalysis data, and the reason for this pattern could be explained by the fact that the historical series of reanalysis has a high consistency in its values, considering the seasonality of the series that facilitates the training process, leading to a better optimization of the error in the network. The results for buoy location five also have an accuracy above 90% for all lead times, and the same quality of the historical series is observed.

[1] Here, the accuracy is obtained subtracting MAPE from 100%.

Some disparity is seen between predictions and observed values by the buoys at locations five and seven. As we can see in Figs. 6 and 7, the buoy data have a higher variance from the mean, with values ranging from low heights to higher heights, but not crossing 2m. This leads the LSTM (and also the reanalysis) to smooth the prediction around a mean value. Nevertheless, the pattern of the observation data is well described by the deep learning algorithm. This behaviour is not seen in buoy location number two, and a possible reason for that being due to the difference in water depths in the regions where the data are obtained. Buoys locations number five and seven are in shallow water while buoy location number two is in deeper water.

The predicted results for buoys locations number one, three, four and six are shown in Figs. 8 and 9. Predictions for buoys locations number one and three follow closely the data from reanalysis and observations, while despite the fact that reanalysis and predictions are very similar, a little discrepancy is seen between buoys locations four and six, if compared to observations. This could also be due to the depth of the water in this four buoys locations.

5.2. Multivariate predictions of H_s

So far we have considered historical series of only significant wave height as input to our model. However, we could consider more than one variable as input to train the network, and also consider one (or multiple) output as the prediction. In this sense, the LSTM will capture how the data from other variables can improve the predicted variable. We call the variables that will not be predicted, but used as input to the model as features, while the significant height, in this work, will remain our target variable. In what follows, we decided to conduct our experiments with buoy location number one, since this has a lower global accuracy compared to the others, both for MAPE and MAE. Our multivariate simulation, the architecture of the network and the parameters used are the same as the single variable simulation. The difference here is that two more variables are considered in the input: the peak wave period T_p and the 10m wind speed U_{10}. As output, we have only the prediction of the significant wave height. Note, however, that H_s is also used as input of the model.

We show in Fig. 10 the predicted values of significant wave height for this multivariate version of our algorithm. Metrics are calculated

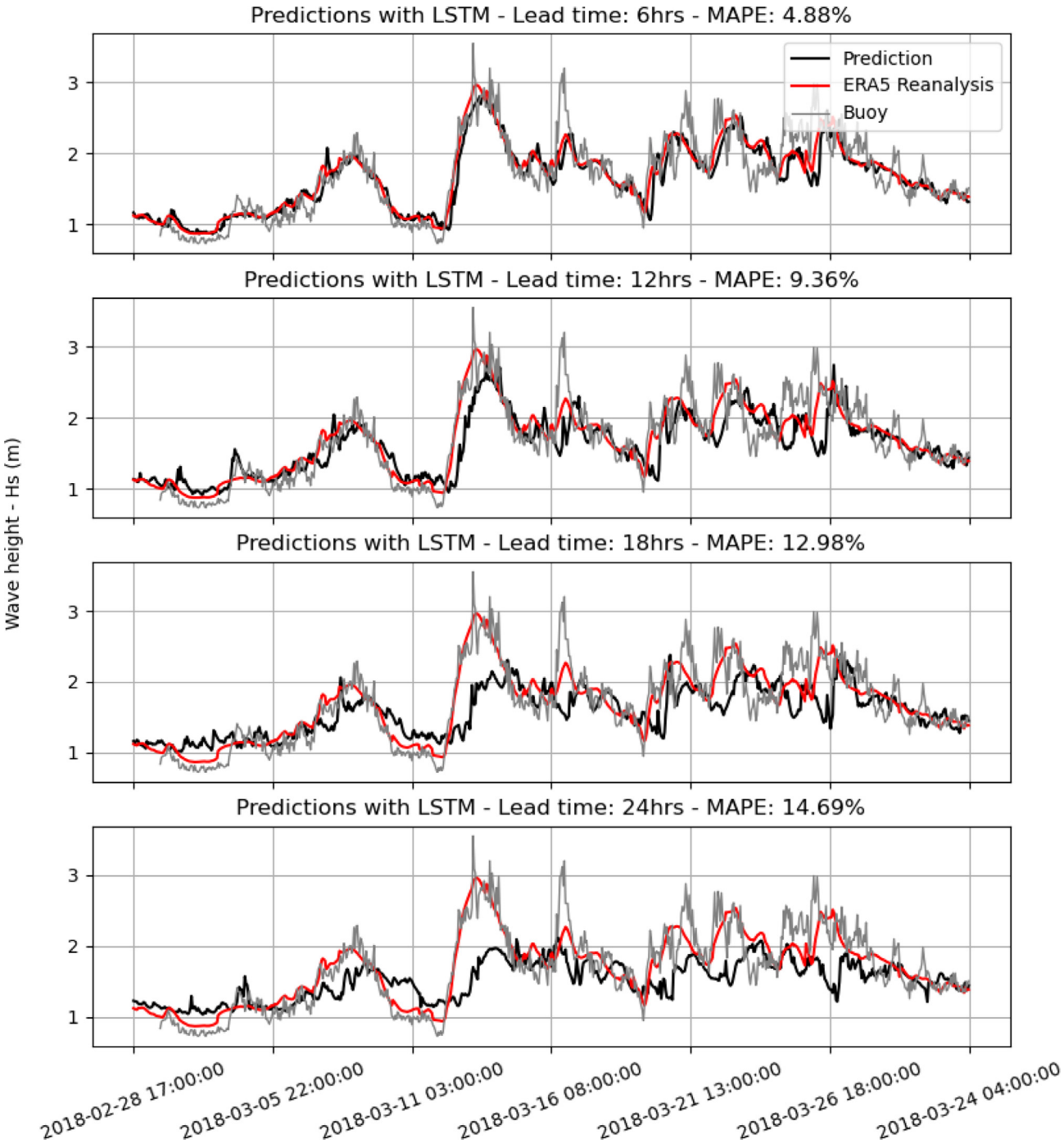


Fig. 5. Predictions (black) and reanalysis from ERA5 (red) comparison for buoy location number two. The blue dots show real observations obtained from the buoy.

using ERA5 as reference value. We can see that there is no considerable improvement in the result and the accuracy of the predictions using two more variables as input. In fact, a large error is observed for lead times 12, 18 and 24, and the MAPE for these lead times are similar, as well as the trend in the graphs. We can conclude here that, the gap between the data used in the training and the effective prediction, i.e, the lead time from 12, affects the results and creates a pattern that is in fact the same in the predictions. This behaviour, however, is not seen in the lead time 6, since the gap between the last training data and the prediction is smaller. Thus, based on the accuracy found, we can state that the multi variable simulation has not improved the results from the LSTM network, as already observed in other works in the literature using non-recurrent neural networks (Londhe and Panchang, 2006).

Roughly speaking, in a model based only in data, the best correlated features will not, in theory, provide an improvement of the predictions, due to the fact that it does not aggregate any new information. Features that are physically known to affect the target variable (either linearly or non-linearly) may not improve the accuracy of our model, and can even reduce it comparing to a single variable simulation. This could happen because adding more features as input of your model, the network will try to see the pattern between these values and thus try to correlate then, based on what the historical series show. Therefore, we also

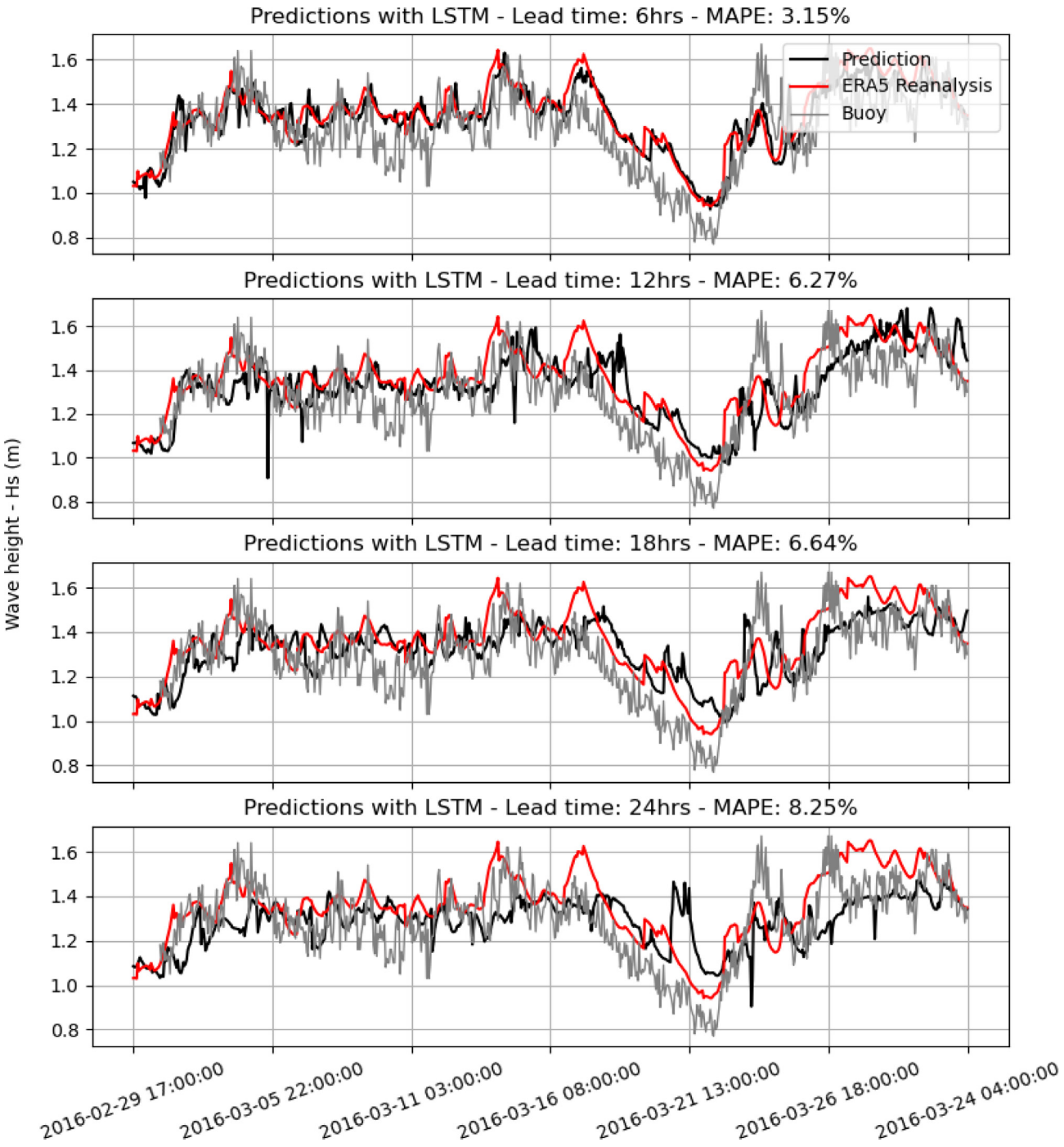


Fig. 6. Predictions (black) and reanalysis from ERA5 (red) comparison for buoy location number five. The grey dashed line show real observations obtained from the buoy.

developed a multivariate analysis using the best correlated variables from those available in ERA5 database with H_s. We use the Pearson's correlation coefficient, where we multiply deviations from the mean for the variable 1 times those for variable 2, and divide by the product of the standard deviations. We have:

$$r_{\text{correlation}} = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{(n-1)s_x s_y}. \tag{9}$$

The correlation coefficient always lies between one (perfect positive correlation) and minus one (perfect negative correlation); zero indicates no correlation. We show in Fig. 11 the correlation between twelve variables available in ERA5 and H_s. It is important to note that this correlation is based only in data, and does not take into consideration the physical relationship between the variables. As maximum individual wave height, altimeter wave height and altimeter corrected wave height are obviously identically correlated ($r_{\text{correlation}} = 1$) and can bring no improvement to the training, we choose the following four variables for our multivariate simulation:

1. Mean wave period based on the second moment for swell;
2. Mean wave period based on the first moment for swell;
3. Mean wave period based on the second moment for wind waves;
4. Mean wave period based on the first moment for wind waves.

We have executed the algorithm with the input training data consisting of five variables: the significant wave height, which is our target

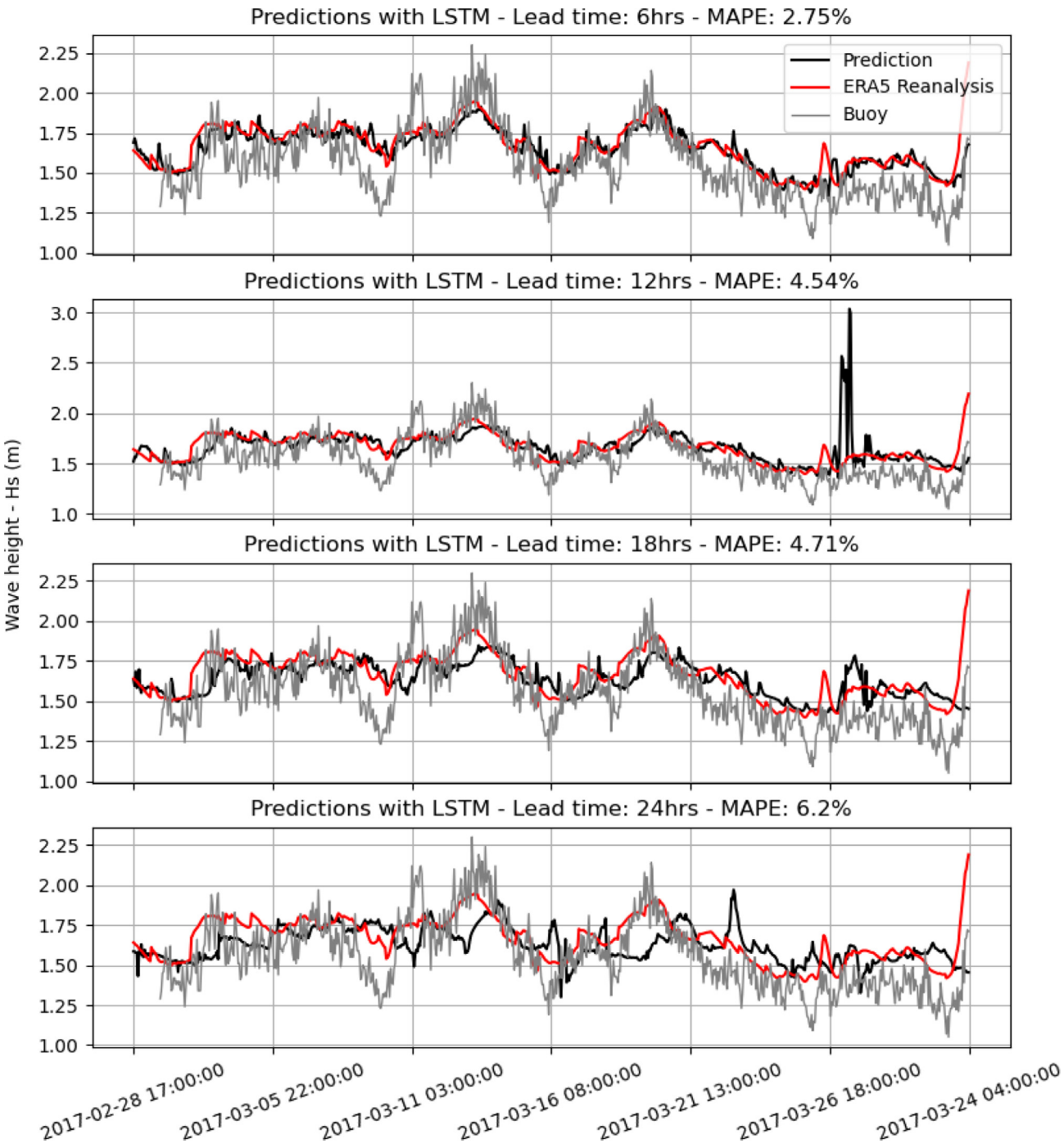


Fig. 7. Predictions (black) and reanalysis from ERA5 (red) comparison for buoy location number seven. The grey dashed line show real observations obtained from the buoy.

variables, and the four enumerated above. The results are shown in Fig. 12 and as we can see, the they do not improve significantly the previous in this subsection.

5.3. Predictions not considering H_s as feature

In this section, we train our model using H_s as the target variable, while peak wave period and the 10m wind speed are used as inputs. Note that this differ from the previous sections, where H_s is also considered as feature. Thus, the weights in the network are updated optimizing the loss function using only those variables and aims to reproduce the target. Again, metrics are calculated using ERA5 as reference value.

With the same goal of improving the results at location number one, we use the time series for this point in space. Fig. 13 shows the results of this framework of training. The accuracy drops with this strategy, although are above 70% for all lead times. These results show the obvious dependency of the physical variables (peak period and wind) with the significant height, since the trend of the series is well described for lead time 6. Naturally, the metrics indicate a deterioration in accuracy for larger lead times. The trend in the series is somewhat lost for lead time 24 on, as the peaks are not reproduced by the LSTM.

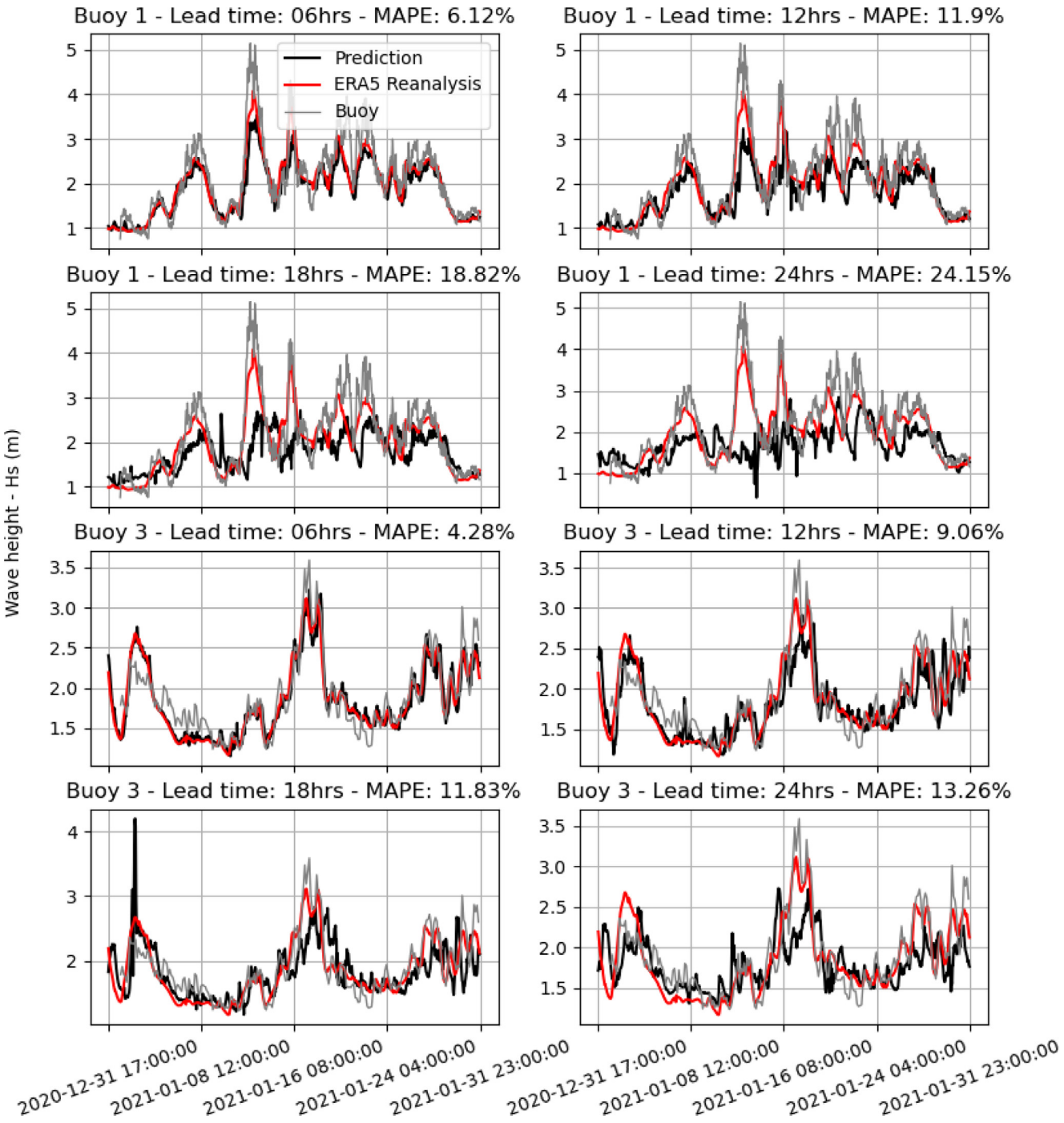


Fig. 8. Predictions (black) and reanalysis from ERA5 (red) comparison for buoys locations number one and three. The blue dots show real observations obtained from the buoy.

5.4. Predictions using real buoy data in the training set

We now present the predictions obtained with buoy measured data as the training set. Only the significant wave height is considered in the set of features and target. Two buoys locations, namely, number 1 and 2, are used for this analysis. Both locations have buoy data available since April/2009, but there are several missing values and outliers, due to buoy issues and maintenance. Buoy location number two has more erroneous measurements compared with buoy location number one, and therefore the k-Nearest Neighbours classification algorithm was applied to filter these outliers, which allowed the removal of 1.6% of data for location number one and 14.5% for location number two. A detailed statistics summary can be found in Bose et al. (2022).

Originally, the data were hourly, but with the outliers removal, gaps in the hourly discretization appeared. Nevertheless, the predictions were made using the same number of points as before, i.e, 751 steps. The verification time period spans from 06-02-2019, at 08 : 00 GMT to 09-03-2019, at 17 : 00 GMT for buoy location number one and from 02-03-2019, at 09 : 00 GMT, to 30-05-2019, at 05 : 00 GMT for buoy location number two. The same architecture and methodology for training of the LSTM network explained previously was used. The metrics are now calculated between predicted values from LSTM and buoy data. Results are presented in Figs. 14 and 15.

It can be observed that the accuracy of the results drop considerably in the predictions using measured data for the algorithm training, instead of ERA5 data. We obtained the MAPE for those two buoy

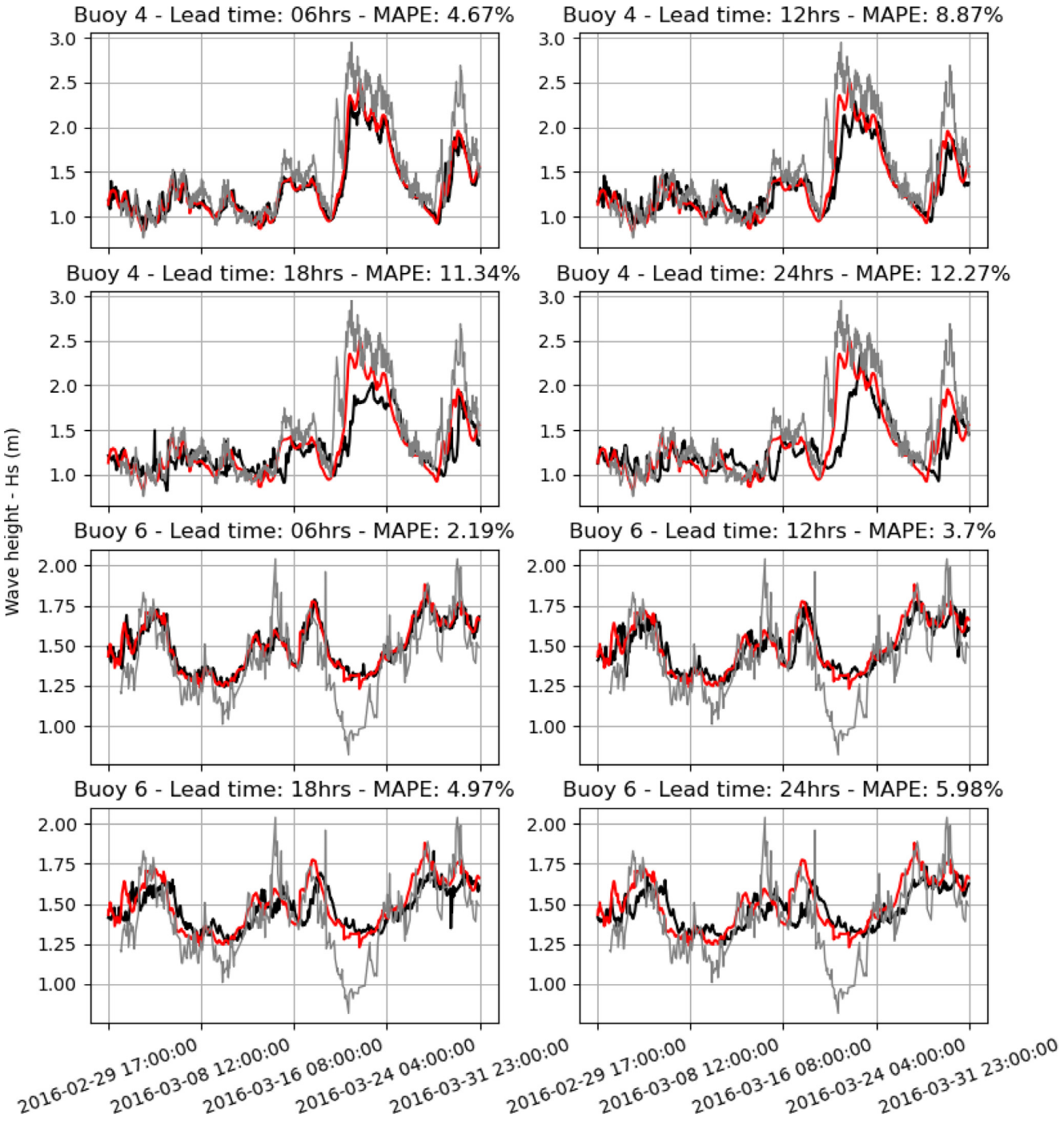


Fig. 9. Predictions (black) and reanalysis from ERA5 (red) comparison for buoys locations number four and six. The grey dashed line show real observations obtained from the buoy.

locations. They are 12.96% (87.04% accuracy) for lead time 6, 19.79% (80.21% accuracy) for lead 12, 22.11% (77.89% accuracy) for lead 18 and 26.17% (73.83% accuracy) for lead 24 for buoy location number one, and a MAPE of 18.26% (81.74% accuracy) for lead time 6, 25.73% (74.27% accuracy) for lead 12, 27.94% (72.06% accuracy) for lead 18 and 29.57% (70.43% accuracy) for lead 24 with buoy location number two. The former has a better global accuracy compared to the latter due to the large amount of data that were filtered out, giving a worst pattern for the historical time series. Since the removal exclude data points in the historical data, both datasets have gaps in the time domain and this information could muddle the network. Furthermore, buoys data, as can be seen by the graphics in Figs. 14 and 15, have a higher variance and noise in the time series.

Comparing the accuracy of our proposed methodology with the accuracy of ERA5 data, against the real buoy data, we have a great difference for buoy location number one and a similar result for buoy location number two. Figs. 14 and 15 show the MAPE metric for ERA5 reanalysis compared to buoy data. Note that ERA5 dataset is not a forecast, differently from LSTM predictions. For lead time 6 h, the LSTM accuracy is 87.04% against 80.67% of ERA5, in location number one.

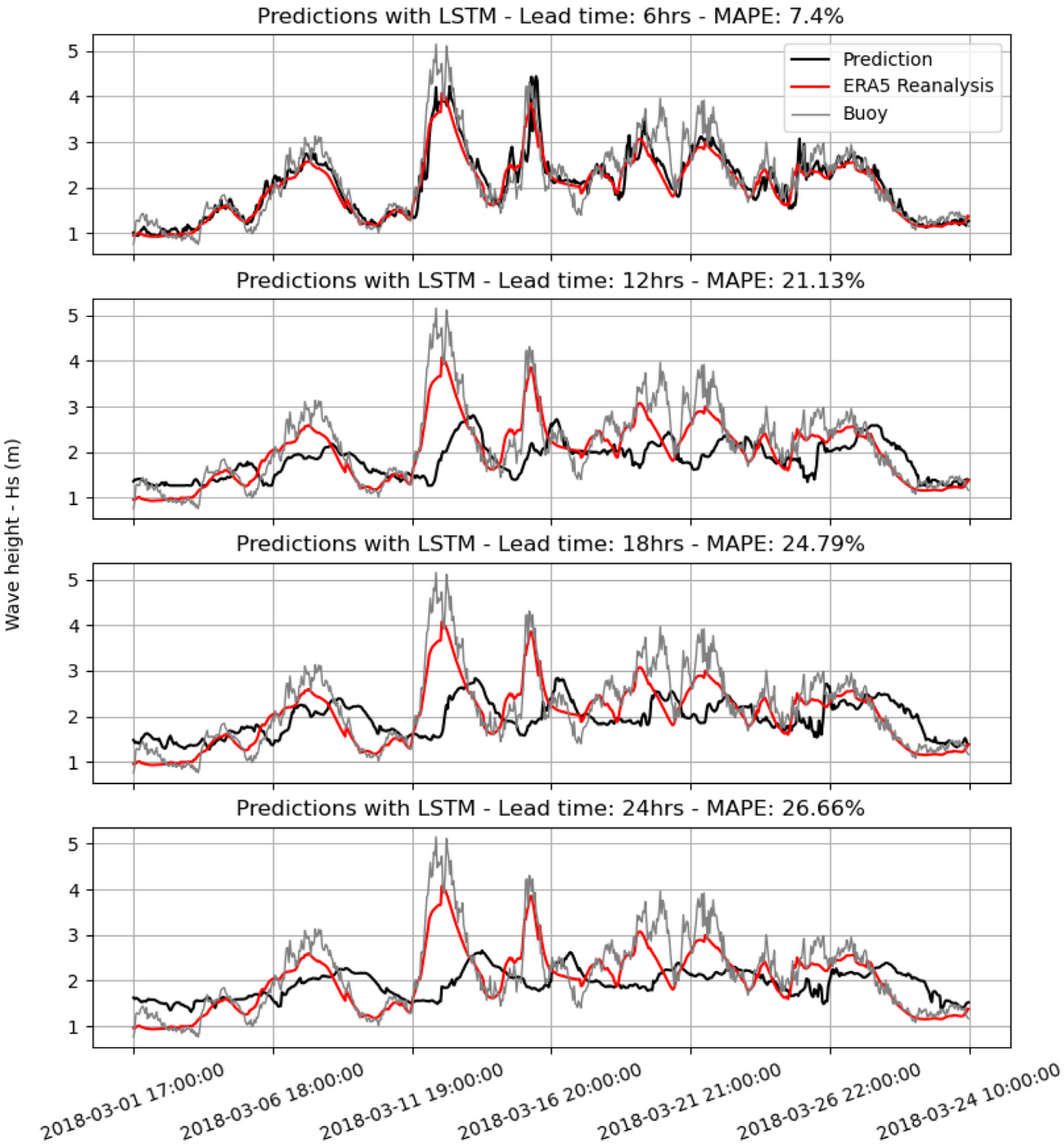


Fig. 10. Predictions (black) for buoy location number one and reanalysis from ERA5 (red) comparison for multi variable (T_p and U_{10}) simulation. The grey dashed line show real observations obtained from the buoy.

For location number two, the difference is small, although still superior for LSTM: 81.74% against 80.59%. Considering that ERA5 data are more accurate than any forecast produced with the physical model on which is based on, we infer that LSTM with the methodology developed in this work can improve the 6 h forecast which uses a physical model.

6. Conclusion

We have presented a deep learning strategy based on long short-term memory to forecast significant wave height in seven different locations on the Brazil's coast, for four lead times. A three layer architecture is built and training is performed for datasets with different sizes, depending on the lead time. Our purpose was to develop a new training methodology for a machine learning technique, that has the ability to preserve the lead time between the last known step and the prediction step. In this sense, ERA5 dataset is an acceptable resource to validate our strategy, and when using buoy data, we can improve what reanalysis accomplishes, with lower computational cost and resources.

The LSTM results show that accuracy, with respect to the reanalysis data, depending on the location, can reach almost 95% for very short-range forecast. For larger lead times, the accuracy naturally decreases, as the gap between the training set and the prediction becomes larger. Nevertheless, acceptable results are obtained in a computational time that is proportional to the size of the training set. For simulations using only buoy data for training, the accuracy of the LSTM model is 87%

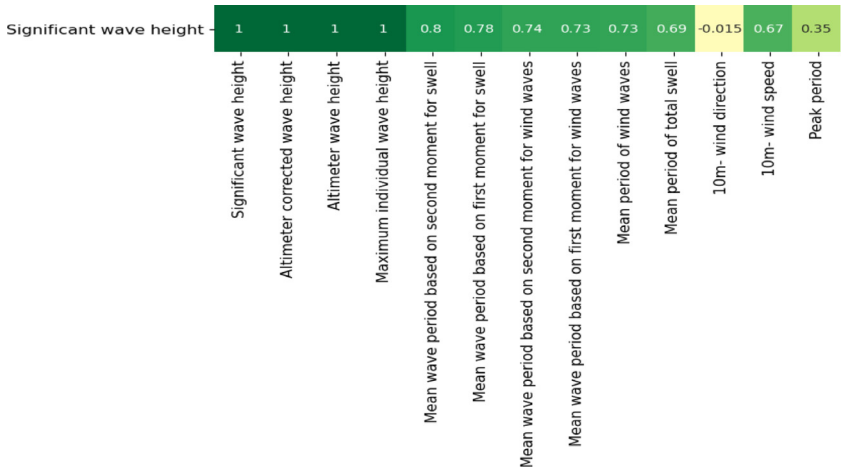


Fig. 11. Pearson's correlation coefficient between the best 12 ERA5 ocean wave variables and H_s.

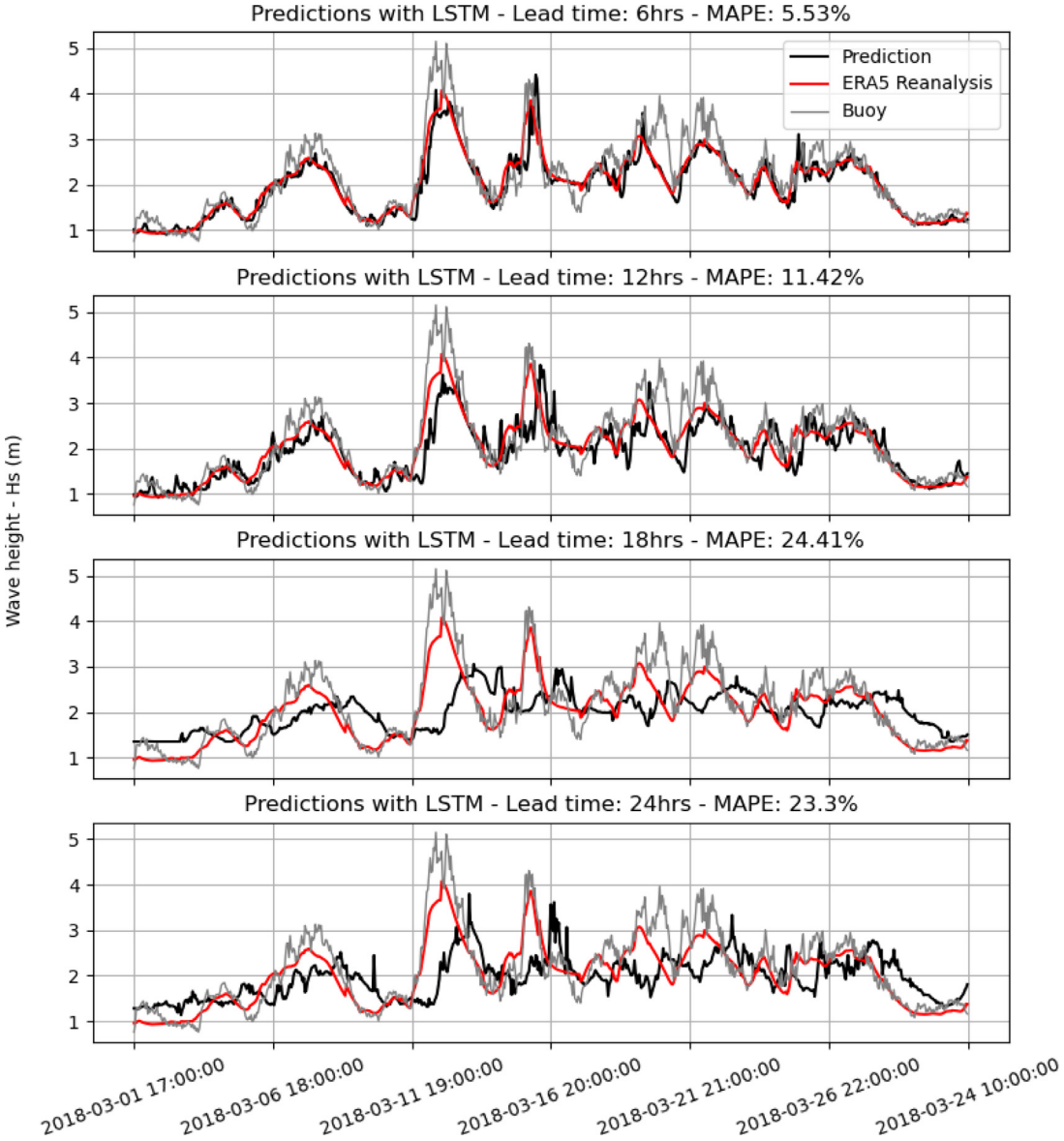


Fig. 12. Multi-variable prediction for buoy location number one using LSTM for the best correlated variables. Predictions are in black, reanalysis from ERA5 in red and buoy data given in grey.

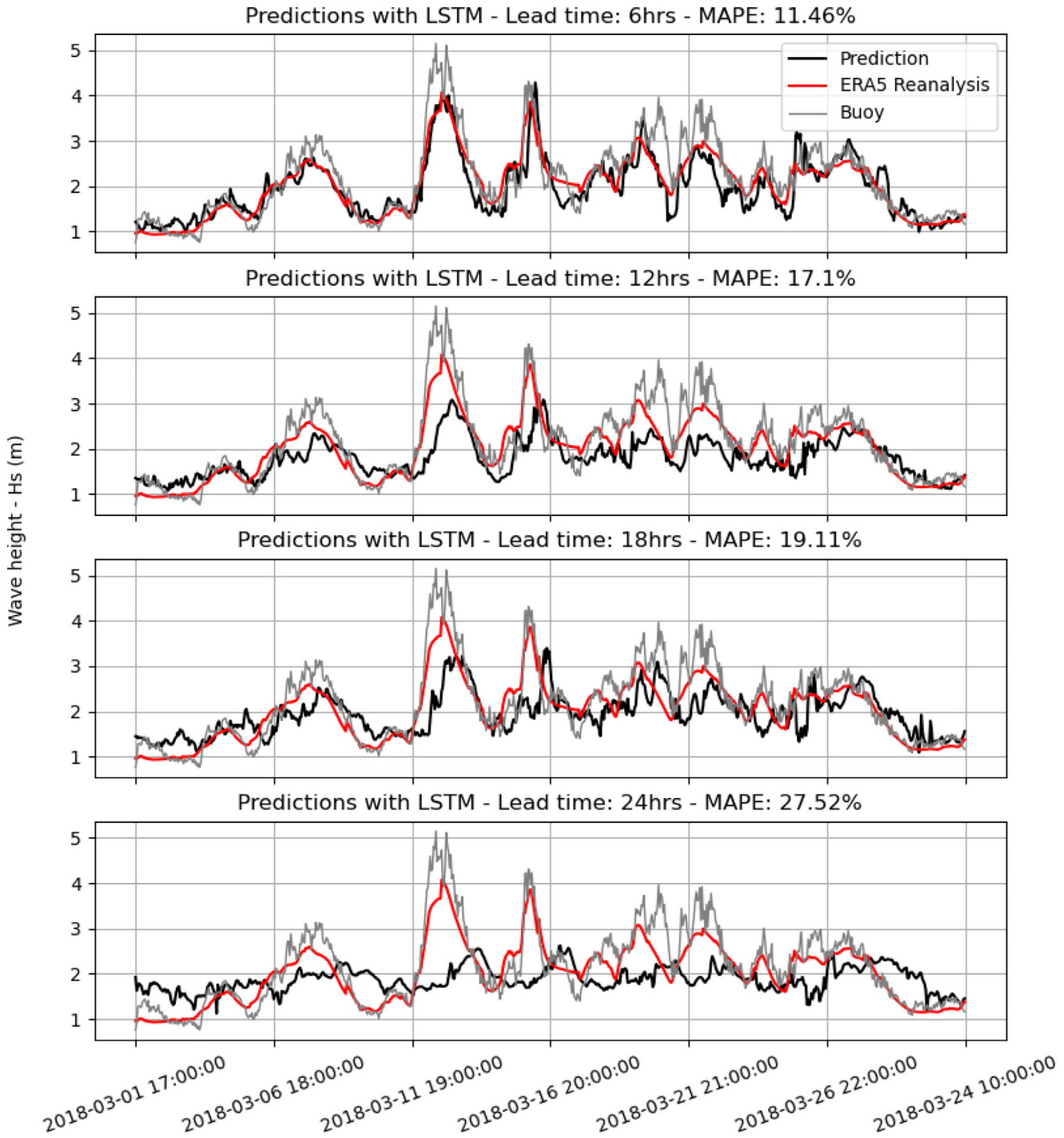


Fig. 13. Prediction using LSTM for buoy location number one considering only T_p and U_{10} as the inputs. Predictions are in black, reanalysis from ERA5 in red and buoy data given in grey.

for lead time of 6 h, and our predictions outperform ERA5 reanalysis results when compared with the observational data, as can be seen in Figs. 14 and 15, for this lead. This demonstrates the ability of the methodology presented here to forecast significant wave height.

The explosion on the number of works and applications which use machine and/or deep learning algorithms is remarkable. Public domain code libraries and the increasing amount of available datasets (ERA5 has recently increased its historical data back to 1958), appears to be an opportunity to improve present ocean wave forecasting. One of the main advantages of machine or deep learning approaches is the computational time needed for obtaining a result. Machine and deep learning models are, nowadays, easy to implement and can be implemented without a profound knowledge of the model's physics.

Despite the very promising advantages of using data-driven models, a better understanding of a number of their operational aspects are necessary, as well as a stronger mathematical foundation is needed. Neural networks, although typically faster than physical models, can have its processing time scaled with data size and regions to be modelled. A global prediction, for every grid point in the globe, considering a 0.5 grid spacing of ERA5, might be not feasible with the method of this paper. Another drawback is the fact that neural networks are still partially "black box" models, i.e., we cannot control the parameters

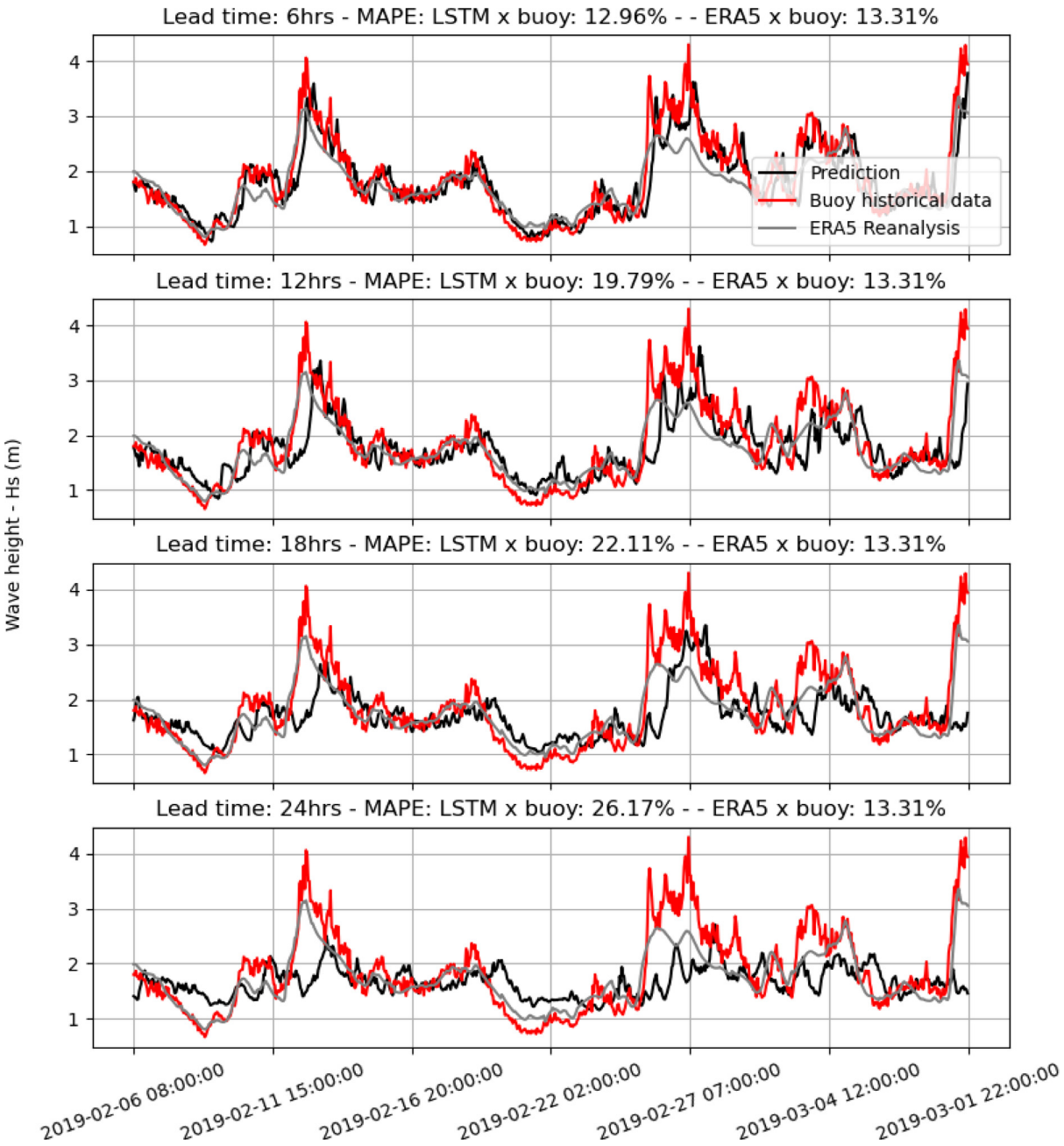


Fig. 14. Prediction of H_s using LSTM for buoy location number one using measured data as feature. Predictions are in black and buoy data given in red.

that are learned within the training phase, making the results difficult to analyse from a physical point of view. Nevertheless, the lack of physical interpretation is a deficiency that can be solved through physical informed neural networks.

Approaches different from the ones used in this work can (and should) be tested. Our strategy was based both on complexity of the network and computational time. In this sense, further studies are necessary to contemplate this goal.

CRediT authorship contribution statement

Felipe C. Minuzzi: Investigation, Conceptualization, Methodology, Software, Writing – original draft. **Leandro Farina:** Conceptualization, Methodology, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Felipe Minuzzi reports financial support was provided by Office of Naval Research. Leandro Farina reports financial support was provided by Coordination of Higher Education Personnel Improvement.

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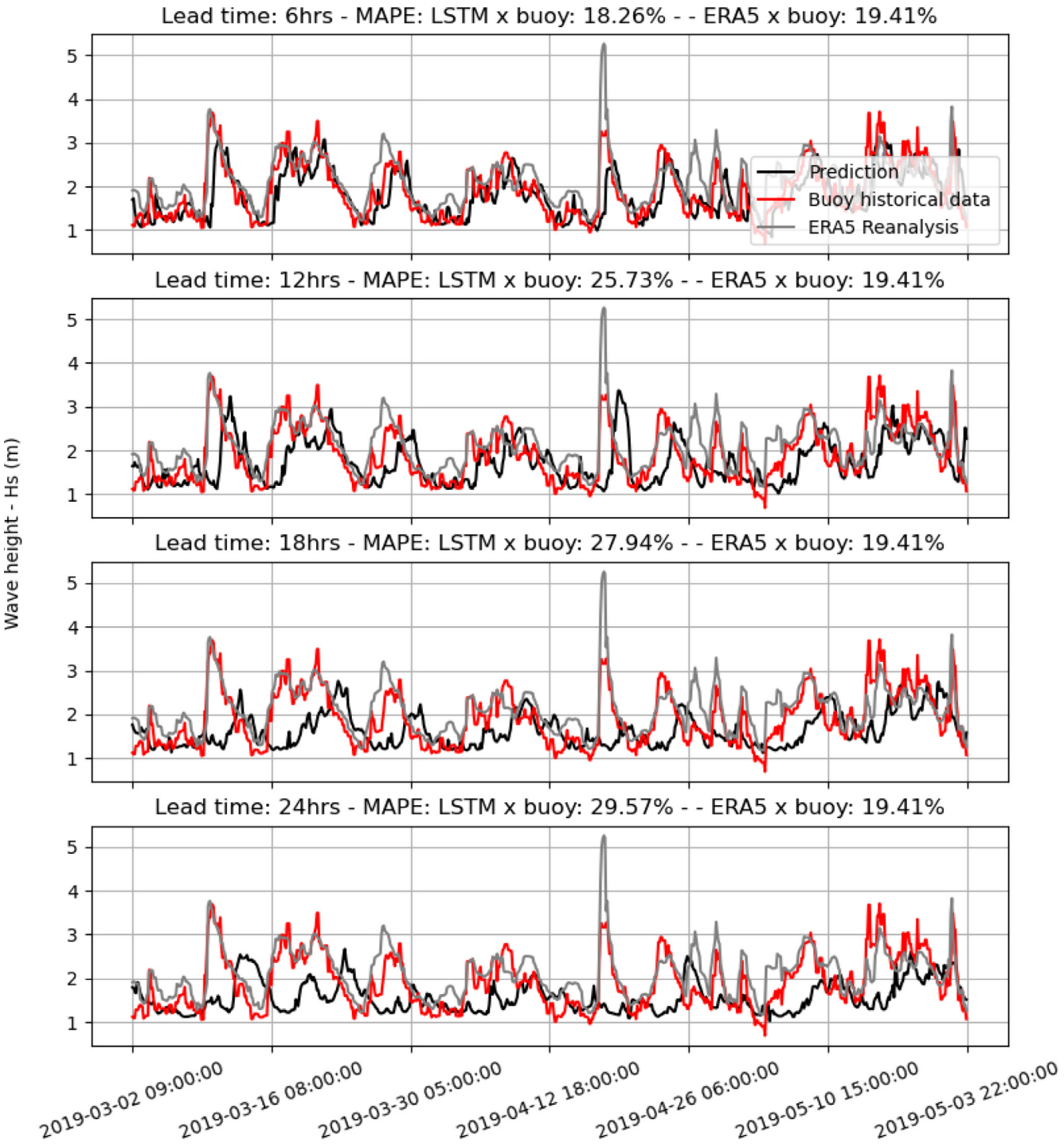


Fig. 15. Prediction of H_s using LSTM for buoy location number two using measured data as feature. Predictions are in black and buoy data given in red.

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